

DEEP PYQ ANALYSIS

GATE DA 2027

High-Probability Question Bank
& 10 Full-Length Mock Tests



Marks per section · three counted papers

MD ANISH AHAMAD

GATE DA 2027 — Deep PYQ Analysis & High-Probability Question Bank

By MD ANISH AHAMAD · First Edition 2026

This preview contains 34 real, unedited pages from the 614-page book, exactly as they appear in the full PDF. Page numbers in the running heads and on the contents pages refer to the full book.

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The full book also has: all eight subject chapters, 257 practice questions with worked solutions and four confidence figures each, all ten mock tests with keys and solutions, the hundred to work first, the what-not-to-skip chapter, the revision plan and the full last-minute sheet.

GATE DA 2027

Deep PYQ Analysis & High-Probability Question Bank

*Every GATE DA paper ever set, counted question by question, and 907 newly written questions built on what they show
for the GATE Data Science and Artificial Intelligence paper*

MD ANISH AHAMAD

First Edition · 2026

For the examination of February 2027

gateguide.co.in

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Preface

Every GATE book tells you what is in the syllabus. The syllabus is one page long and the organisers publish it free. This book tries to answer the question the syllabus cannot: **of the eighty-five technical marks in the GATE Data Science and Artificial Intelligence paper, where do they actually land — and how much of that can anyone honestly claim to know?**

For this paper the second half of that question matters more than usual. **GATE first set a Data Science and Artificial Intelligence paper in 2024.** The whole history of the examination is three papers — 2024, 2025 and 2026 — one a year, each set by a different institute. The organisers have never published a marks breakdown, so every "subject-wise weightage" table you have seen for DA is somebody's count of at most three papers, and usually it does not say where it put the questions that sit on a boundary between two sections. In a paper whose sections overlap as much as this one's do — probability inside machine learning, eigenvectors inside principal component analysis, graph search inside AI — that is not a small omission.

So this book does not use those tables. It counts the papers. All three of them.

How this book was built

Three papers, counted question by question. Every GATE DA paper there has been — 2024, 2025 and 2026, 195 questions and 300 marks — was read one question at a time. Each question was assigned to a section of the 2027 syllabus and to a topic inside it, and its marks and question type were taken from that year's official answer key, never from the paper's layout. Each paper was made to close: 65 questions, 100 marks, 30 at one mark and 35 at two, General Aptitude at exactly 15. A count that does not close is a count with a mistake in it, and all three close.

Every citation names a year and a question number. Not once in this book will you find a claim like "this pattern keeps recurring". DA is set as one paper a year, so a citation such as "2025 (Q35)" names exactly one question, and every historical statement in this book can be checked against a paper you can download from an organising institute's own website.

Every boundary decision is written down. Twenty-three of the 165 technical questions could honestly have been counted in a second section. Each records the section it was counted in, the section it could have gone to, and why, so that a reader who disagrees can move it and see exactly what changes. Six were not settled by any stated rule, and each of those is marked as counted at its lean rather than presented as certain.

The 2027 syllabus was compared with 2026, word by word, from the two official PDFs, by a program rather than by reading. The result is short and, for once, simple: **no phrase added, no phrase removed, no phrase reworded.** The only differences are labels — the 2027 document numbers its seven sections, and drops the "(i)" and "(ii)" in front of Supervised and Unsupervised Learning. Section 2 prints the comparison and the method. A change list that is wrong is worse than none, and a change list that says "nothing changed" is the one most likely to be doubted, so it is the one most worth generating from the source rather than asserting.

Every number in every question was computed, not remembered. Probabilities were evaluated, eigenvalues and singular values were found, SQL results and relational-algebra outputs were worked row by row, search trees were expanded node by node, and every fragment of Python was written to be run and its output taken from running it. Every incorrect option is the output of a specific mistake, and the solution names that mistake, so that a candidate who chose it learns what they did rather than merely that they were wrong.

What this book refuses to do

It does not claim to know the paper. Four commitments follow:

- **Facts, analysis and predictions are labelled separately.** A statement about what happened in 2025 is marked FACT. A statement about what the data suggests is marked ANALYSIS. A statement about 2027 is marked PREDICTION. A statement that is the author's opinion is marked JUDGEMENT. They are never blended.
- **Confidence is reported in four dimensions, not one.** Being confident that Bayes' theorem will be examined is not the same as being confident about a particular Bayes' theorem question. Every question carries Topic, Concept, Question-pattern and Exact-question confidence, and the last is capped at 35 per cent because GATE repeats concepts, not wording.
- **The confidence figures are called estimates, because that is what they are.** They are not a fitted model and they have not been validated. Three papers could not validate them if anyone tried. Section 7 explains how they were arrived at and what they cannot support.
- **Absence of evidence is printed as absence of evidence.** Several topics named in the 2027 syllabus produced no question at all in three papers. Each is said so plainly, with no history invented to fill the gap.

Two things the counting found that should change how you study

The syllabus has not moved, so the zeros in this book are real zeros. The 2026 paper was set on exactly the text a 2027 candidate will sit, and no question in any of the three papers falls outside it. When a named topic has never been examined, that cannot be explained away as "it was not on the syllabus then". Confidence intervals and the z-, t- and chi-squared tests, LU decomposition and linked lists are all named in the 2027 syllabus, and **none of the 165 counted technical questions was counted to any of them.** That is not evidence that they will be skipped again — three papers are too few for that — but it is a fact that no previous-year practice can cover them, and this book writes questions on them on the authority of the syllabus alone and says so.

Two sections have already swung by nine marks or more. Calculus and Optimization carried 8 marks in 2024, 11 in 2025 and **2** in 2026. Database Management and Warehousing carried **7**, 11 and 18. Over three papers the largest technical section has been Programming, Data Structures and Algorithms once and Probability and Statistics twice. A candidate who plans on any one year's shape is planning on one draw from a range that three papers have only begun to show. This book prints the range beside every mean, and the range is the number to plan on.

The two findings point the same way: in a paper this young, the syllabus is at least as good a guide to what can be asked as the past papers are, and the past papers are best used to learn how a topic is asked rather than whether it will be.

What is in it

The official 2027 syllabus with the 2026-to-2027 comparison generated from the source documents; a chapter on the organising institute and why no DA paper yet carries its name; the counted weightage of every section and, inside each section, of every topic, over all three papers; eight subject chapters — one for each of the seven technical sections and one for General Aptitude — with 257 newly written questions, each with a worked solution and four confidence figures; a chapter explaining exactly what those figures mean and what they do not; a hundred-question priority list; a chapter on what not to skip; ten full-length mock tests in the 2027 pattern with keys and worked solutions, 650 questions more; a staged revision plan; a last-minute sheet; and a statement of sources and their limits.

Every one of the 907 questions is new. None is copied from a past paper, and the mock tests were written under a rule that forbids reusing any question from the subject chapters. What is borrowed from the past is the pattern, and the pattern is what repeats.

Section 1 — Executive Summary

If you read nothing else in this book, read this. Everything here is drawn from the three official GATE Data Science and Artificial Intelligence papers — every DA paper there has been — counted question by question against their official answer keys, and from the two official syllabus documents. Every figure can be checked; every claim names what it rests on.

Statements are labelled. **FACT** is something counted or quoted. **ANALYSIS** is an inference from it. **PREDICTION** is a statement about a paper that has not been set. They are never blended.

The eight things that matter most

1. The 2027 syllabus did not change a single examinable word

FACT. The 2026 and 2027 DA syllabus documents were compared word by word by a program. Result: **0 phrases added, 0 removed, 0 reworded**, across all seven sections and all 132 phrases. Not one word appears in either document that is missing from the other. The only differences are labels: 2027 numbers its sections "Section 1:" to "Section 7:", and drops the "(i)" and "(ii)" in front of Supervised and Unsupervised Learning.

ANALYSIS. This is the most useful fact in the book, and the least exciting. **The 2026 paper was set on exactly the text you will sit, and no question in any of the three papers falls outside it.** Nothing in the record has to be discounted as deleted material, and no topic's absence from 2026 can be excused as "not on the syllabus then". (The 2024 and 2025 syllabi name the same phrases word for word as well; 2024's differs from 2026's by a single dash, Section 2.3.)

What to do. Nothing you studied against the 2026 syllabus has been removed, and nothing new has been added that you must find time for. Verify the claim yourself: the two documents are one page each.

2. Probability and Statistics is the largest section, and it has grown in both years

FACT. Probability and Statistics carried **15, 19 and 21 marks** in 2024, 2025 and 2026: a mean of **18.3**, and **55 of the 255** technical marks set so far, 21.6 per cent. It was the largest technical section in 2025 and 2026; in 2024 Programming, Data Structures and Algorithms led it, 20 to 15.

FACT. Six further questions, **8 marks**, counted in Machine Learning, AI or Database Management because they ask about a method — naive Bayes, a Bayesian network, a decision-tree split, z-score normalisation — could equally have been counted here.

ANALYSIS. Two rises in a row is not a trend when there are only two steps to look at. What the count does support is simpler: this is the section a DA candidate can least afford to be weak in, because its methods decide questions in three other sections as well.

3. General Aptitude is 15 marks in every paper, and every question in it has been an MCQ

FACT. Ten questions — five at one mark, five at two — totalling exactly 15 marks, in all three papers. All thirty General Aptitude questions set so far were multiple-choice questions.

ANALYSIS. The only block of the paper with no variation at all. It is worth more than any technical section except Probability and Statistics and Programming, Data Structures and Algorithms.

What to do. Do not leave it to the last month. It is the most predictable 15 per cent of your score.

4. IIT Madras sets 2027, and it has never set a DA paper

FACT. GATE 2027 is organised by IIT Madras. The three DA papers were organised by **IISc Bengaluru (2024), IIT Roorkee (2025) and IIT Guwahati (2026).** IIT Madras last organised GATE in 2019, five years before the DA paper existed.

ANALYSIS. There is no institute-matched evidence for this paper at all. Any claim that "IIT Madras prefers" some kind of DA question is not an inference from anything; it is invented. The one public decision IIT Madras has made about DA is the syllabus — and it left every examinable word as it was.

5. Three papers show patterns, not trends

FACT. Summing the section-by-section differences, the 2025 paper differs from 2024 by 26 technical marks, 2026 from 2025 by 26, and 2026 from 2024 by 34 — out of 85.

ANALYSIS. The paper's shape moves a great deal from year to year, and there are only three years.

If the three years were shuffled at random, a section would rise in both steps or fall in both steps one time in three; across seven sections, that is two or three sections expected to look like a "trend" by chance alone. The record shows two — Probability and Statistics, and Database Management and Warehousing.

Nothing in this book is claimed as a trend. Where a section moved in one direction twice, it says so and says that three papers cannot tell that apart from chance.

What to do. Read the range beside every mean in this book. Your paper gets one value out of that range and you do not choose which.

6. Where the marks are

FACT. Marks by section in the three papers, counted against the 2027 syllabus, with the mean and range:

Section	2024	2025	2026	Mean	Range
1. Probability and Statistics	15	19	21	18.3	15-21
4. Programming, Data Structures and Algorithms	20	14	14	16.0	14-20
General Aptitude	15	15	15	15.0	15-15
6. Machine Learning	15	12	14	13.7	12-15
5. Database Management and Warehousing	7	11	18	12.0	7-18
2. Linear Algebra	10	12	8	10.0	8-12
7. AI	10	6	8	8.0	6-10
3. Calculus and Optimization	8	11	2	7.0	2-11

ANALYSIS. The top three technical sections — Probability and Statistics, Programming, and Machine Learning — carry 144 of the 255 technical marks, 48.0 a paper. Machine Learning is the steadiest technical section (12 to 15). **Database Management and Warehousing and Calculus and Optimization are the least predictable blocks in the paper:** one has run 7, 11, 18 and the other 8, 11, 2.

7. Most topics have been examined more than once — and four named topics never

FACT. Grouping each section's syllabus into the topic groups its chapter uses gives 52 groups. **21 appeared in all three papers, carrying 155 of the 255** technical marks; 19 appeared in two papers (88 marks); 8 in one (12 marks); and 4 in none. The four are confidence intervals with the z-, t- and chi-squared tests; LU decomposition; linked lists; and functions of a single variable treated as a topic on their own.

ANALYSIS. Three-fifths of the marks set so far sit in topics every paper has used. That is the core to secure first. The four never-examined groups were named, in the same words, in the syllabus the 2026 paper was set on, so their zero is a real zero — and it still says nothing about 2027.

What to do. Study the all-three-papers topics to depth, and give the never-examined topics a short, honest reading: they are named, they are short, and no past paper can prepare you for them.

8. The multiple-select question is a quarter of the technical paper

FACT. MCQ, MSQ and NAT counts over the full 65 questions: **37, 12, 16** in 2024; **35, 18, 12** in 2025; **33, 14, 18** in 2026. Every MSQ and every NAT was in a technical section. Across the three papers, **44 of the 165 technical questions** were MSQs.

FACT. Three MSQs had **exactly one correct option** in the official key: 2025 Q27, 2026 Q23 and 2026 Q51.

FACT. MSQ carries no negative marking and no partial credit. You score only by selecting every correct option and no incorrect one.

ANALYSIS. DA has had MSQs from its first paper; there has never been a DA paper without them. Prepare for between roughly 12 and 18 in your own paper, and practise judging each option as a separate true-or- false statement — including the possibility that only one is true.

What this book will not do

It will not tell you it knows the paper. Three commitments follow from that, and they are kept everywhere in this book:

- **Facts, analysis and predictions are labelled separately**, as above.
- **Confidence is reported in four dimensions.** Being confident that principal component analysis will be examined is not the same as being confident about a particular PCA question. Every question in the bank carries Topic, Concept, Question-pattern and Exact-question confidence, and the last is capped, because GATE repeats concepts and not wording.
- **Absence of evidence is printed as absence of evidence.** Several topics named in the 2027 syllabus produced no question in three papers. Each is said so plainly, with no history invented to fill the gap, and questions written on syllabus authority alone say that this is what they rest on.

It will not cite a question it has not counted. Every historical claim in this book names the year and the question number — "2026 (Q34)" — and DA sets one paper a year, so that citation identifies exactly one question in an official paper you can download.

It will not copy a question. Every question in this book was written for it, and every numerical answer was computed rather than estimated.

Where the count needed a judgement, and how it is shown

DA's sections overlap more than most papers' do, and the count had to decide where each overlapping question belongs. The rules are stated once, in Section 2.4, and applied to all three papers: a question about what a machine-learning method does goes to Machine Learning even when its arithmetic is probability or linear algebra; a question whose answer is the probability or matrix fact itself goes to that section;

Bayesian networks and AI-framed search go to AI; plain graph traversal goes to Programming.

2.2 What changed between 2026 and 2027

FACT. The comparison below was made by a program that reads the two official PDFs and compares them phrase by phrase and word by word. No coaching site's "GATE 2027 syllabus changes" article was opened, quoted or checked against: a change list copied from another change list is worth nothing, and a wrong one is worse than none.

The method. Each section's text is split into phrases on its own commas, semicolons, colons and full stops. The 2026 and 2027 phrase lists are aligned after folding only case and spacing, so that a change of a single word or letter would appear as a reworded phrase. Separately, each section is compared token by token — every word and every punctuation mark — which is what finds a change such as the removal of a list marker, which a phrase split cannot see. Finally the whole of each document is checked for any word that appears in one and nowhere in the other.

The result.

Section	2026 phrases	2027 phrases	Added	Removed	Reworded	Labelling only
1. Probability and Statistics	35	35	0	0	0	0
2. Linear Algebra	18	18	0	0	0	0
3. Calculus and Optimization	6	6	0	0	0	0
4. Programming, Data Structures and Algorithms	18	18	0	0	0	0
5. Database Management and Warehousing	19	19	0	0	0	0
6. Machine Learning	26	26	0	0	0	2
7. AI	10	10	0	0	0	0
Total	132	132	0	0	0	2

Whole-document vocabulary check. Words in the 2026 syllabus that appear nowhere in the 2027 syllabus: none. Words in the 2027 syllabus that appear nowhere in the 2026 syllabus: none.

Six of the seven sections are identical word for word and punctuation mark for punctuation mark. The two entries in the last column are both in Section 6, and both are labels:

	2026	2027
Section 6, first paragraph	"(i) Supervised Learning:"	"Supervised Learning:"
Section 6, second paragraph	"(ii) Unsupervised Learning:"	"Unsupervised Learning:", now set as a separate paragraph

The layout changed in two further ways, neither touching what is named. The page header now reads "GATE 2027" and "IIT Madras | Organizing Institute", where 2026's read "GATE 2026 IIT Guwahati | Organizing Institute". And the 2026 document opened each paragraph with the section's name and a colon, unnumbered — "Probability and Statistics:", "AI:" — where 2027 numbers the sections "Section 1:" to "Section 7:" and sets each name on its own line. This book's section numbers are the 2027 document's own.

The two documents, for the record.

	2026	2027
Published by	IIT Guwahati	IIT Madras
Pages	1	1
Creation date in the PDF's metadata	2025-08-07	2026-07-06
Phrases	132	132
Tokens (words and punctuation marks)	448	446

2.3 What an unchanged syllabus means for how you study

(a) Every zero in the record is a real zero

FACT. All three counted papers were set on the same words as the 2027 syllabus. The official 2024 (IISc Bengaluru) and 2025 (IIT Roorkee) DA syllabi were compared with 2026's in the same way, phrase by phrase (research/syllabus_changes.md). The 2025 text is identical to 2026's word for word and punctuation mark for punctuation mark. The 2024 text differs by one punctuation mark: a hyphen where 2025 and 2026 print a dash, in Section 7's "reasoning under uncertainty topics — conditional independence representation". All three carry the "(i)" and "(ii)" labels that 2027 dropped. The book also claims, and Section 2.5 shows, that no question in any of the three papers tested anything the 2027 document fails to name.

FACT. Four topic groups named in the 2027 syllabus have drawn **no counted question** in any of the three papers:

Section	Named in 2027, never examined in the three counted papers
1 Probability and Statistics	Confidence interval, z-test, t-test, chi-squared test
2 Linear Algebra	LU decomposition
3 Calculus and Optimization	Functions of a single variable , as a topic on its own
4 Programming, Data Structures and Algorithms	Linked lists

FACT. Several narrower phrases are also named in the syllabus and are not the subject of any counted question, as this book tagged the 165 technical questions: multiple linear regression, k-folds cross-validation and top-down hierarchical clustering in Section 6; linear search and mergesort in Section 4; and discretization and compression among Section 5's data transformations. A question can touch a phrase without being tagged with it, so this second list is a guide rather than a proof; the four groups in the table are the firm claim.

ANALYSIS. In a paper whose syllabus had just changed, a zero like these would say little: the topic might simply have been new. Here it cannot be excused that way — not for 2026 at least, whose syllabus named all four in the same words, and whose setters used none of them. That is worth knowing — and it is still not evidence that 2027 will skip them. Three papers is too few for any zero to be a prediction, and the four groups are short, named and cheap to prepare.

PREDICTION. None made. This book writes questions on every never-examined group, says in each question's tag that it rests on the syllabus alone, and gives those questions lower confidence figures than topics the papers have used.

(b) Every swing in the record happened on an unchanged syllabus

FACT. Between 2025 and 2026, Calculus and Optimization fell from 11 marks to 2 and Database Management and Warehousing rose from 11 to 18 (Section 4.3).

ANALYSIS. Nothing in the syllabus explains either movement: the 2025 and 2026 papers fall entirely inside the same named topics, and 2026's syllabus is 2027's word for word. A swing of that size was therefore the setters' choice from the same menu, and the next setting panel — at a different institute again — makes its own choice from the same menu. That is the plainest reason in this book for planning on the range of each section rather than on its latest value.

Section 3 — The Organising Institute, and the DA Paper It Has Not Yet Set

GATE 2027 is organised by IIT Madras.

Each year one institute takes responsibility for the examination, and the paper is set under its supervision. The Data Science and Artificial Intelligence paper has been set three times, by three different institutes, and **none of them was IIT Madras**. There is no DA paper in existence whose setter is the setter of the paper you are preparing for.

This chapter says so, says what that rules out, and then says what the three papers set by other institutes do show — the features of the paper that survived three changes of organiser. It is a short chapter because the honest version of it is short. A chapter that found an "IIT Madras style" in a paper IIT Madras has never set would be inventing it.

3.1 What is fact here, and what is not

FACT. GATE 2027 is organised by IIT Madras. The official 2027 DA syllabus document carries the header "IIT Madras | Organizing Institute", and the official portal is gate2027.iitm.ac.in.

FACT. The organising institutes of the three counted DA papers are: **IISc Bengaluru (2024), IIT Roorkee (2025) and IIT Guwahati (2026).**

FACT. IIT Madras last organised GATE in 2019, five years before the first DA paper was set.

NOT A FACT, and this book will not pretend otherwise. That an institute "has a style" which carries over into a paper it has never set, or across a gap of years and a change of everyone involved. Papers are set by subject panels, not by buildings. Whatever IIT Madras's panels did in 2019, they did it for papers other than DA, on syllabi other than this one.

3.2 Three papers, three institutes

FACT. Technical marks by section, each paper counted question by question against its official key:

Section	2024, IISc Bengaluru	2025, IIT Roorkee	2026, IIT Guwahati	Mean
1. Probability and Statistics	15	19	21	18.3
2. Linear Algebra	10	12	8	10.0
3. Calculus and Optimization	8	11	2	7.0
4. Programming, Data Structures and Algorithms	20	14	14	16.0
5. Database Management and Warehousing	7	11	18	12.0
6. Machine Learning	15	12	14	13.7
7. AI	10	6	8	8.0
General Aptitude	15	15	15	15.0

ANALYSIS. In DA the organiser and the year are the same variable. Each institute set exactly one paper, in exactly one year, so there is no way to tell whether 2024's 20 marks of Programming reflect IISc

Bengaluru's panel, the first year of a new paper, or neither. The same is true of 2026's 18 marks of

Database Management and 2 of Calculus. **Any claim about what a particular institute "does" in DA rests on one paper and cannot be separated from chance.** This book makes none.

3.3 What IIT Madras has already decided about DA

FACT. The one DA document IIT Madras has published so far is the 2027 syllabus. Its PDF metadata gives a creation date of 6 July 2026. Compared word by word with the 2026 document published by IIT Guwahati, it adds no phrase, removes no phrase and rewords no phrase. What it changes is presentation: it numbers the seven sections, sets each name on its own line, drops the "(i)" and "(ii)" in front of Supervised and Unsupervised Learning, and replaces the page header with its own (Section 2.2).

ANALYSIS. That is the only public decision about the DA paper that can be attributed to the 2027 organiser, and it was a decision to leave the examinable content exactly as it was. It supports planning on the 2026 syllabus and the three counted papers without adjustment. It supports nothing more specific than that.

3.4 What survived three changes of organiser

If one institute's paper cannot tell you about IIT Madras, three institutes' papers can at least tell you what did not change when the institute did.

FACT. In all three papers, set by three different institutes:

Feature	2024	2025	2026
Questions / marks	65 / 100	65 / 100	65 / 100
One-mark / two-mark questions	30 / 35	30 / 35	30 / 35
General Aptitude	10 questions, 15 marks, all MCQ	10 questions, 15 marks, all MCQ	10 questions, 15 marks, all MCQ
Block layout	Q1–5 and Q11–35 one mark; Q6–10 and Q36–65 two marks	the same	the same
MCQ / MSQ / NAT	37 / 12 / 16	35 / 18 / 12	33 / 14 / 18
Section with most technical marks	Programming (20)	Probability and Statistics (19)	Probability and Statistics (21)
Every technical section examined	yes	yes	yes

ANALYSIS. The paper's frame — its size, its mark blocks, its General Aptitude block, and the presence of all three question types — held under three organisers without exception. Those are the safest things in this book to plan on. What moved with the organiser, or with the year, or with chance, is how the 85 technical marks were divided: the three papers differ from one another by 26 to 34 marks, summed section by section (Section 4.4).

FACT. DA has contained multiple-select questions from its first paper: 12, 18 and 14 of them. MSQ carries no negative marking and no partial credit: you score if and only if you select every correct option and no incorrect one. Three of the 44 counted MSQs — 2025 (Q27), 2026 (Q23) and 2026 (Q51) — had exactly one correct option in the official key.

ANALYSIS. Unlike papers that have existed for longer, DA offers no "before MSQ" period to discount. The format is part of the paper as every organiser has set it, and a candidate should expect it in 2027 in roughly the same number, without planning around any particular count.

Section 4 — What Three Papers Actually Contain

This is the chapter the rest of the book rests on. Everything after it — the subject chapters, the question bank, the mock tests — is built on the counts below, so this chapter shows you the counts, the method that produced them, and the places where the method needed a judgement.

It is also the chapter that has to say most plainly how little three papers can prove. **GATE first set a Data Science and Artificial Intelligence paper in 2024. The entire record is three papers, one a year.** Three papers show patterns — what has been asked, how often, in what form. They do not show trends, and this chapter does not claim any.

4.1 How the counting was done

FACT: All three official GATE Data Science and Artificial Intelligence papers were read one question at a time: 2024, 2025 and 2026, **195 questions and 300 marks**.

Each question was read from the organising institute's own question paper — not from a key, not from a summary, not from a solved-papers book. Its marks, its question type and its answer were then taken from **that year's official answer key** and from nothing else. So when this book says a question carried two marks and was a multiple-select question, that is the organisers' record of it and not a reading of the paper's layout.

Each question was then assigned to a section of the **2027** syllabus, and to a topic inside that section, in the syllabus's own words wherever the syllabus has words for it. That is the point of the exercise: a 2027 candidate needs to know where the marks would land under the syllabus they will actually sit.

Every count had to close, or it was not a count. Each of the three closes on:

Check	Required	Result
Questions	65	65 in all three
Total marks	100	100 in all three
One-mark questions	30	30 in all three
Two-mark questions	35	35 in all three
General Aptitude	exactly 15	15 in all three (10 questions)
Technical	exactly 85	85 in all three
Marks and type of every row	as in the official key	match in all three

A count that did not close would have been re-done rather than adjusted. Nothing was moved, merged or dropped to make a total come out right.

All three papers use the same layout, and the count confirms it question by question: Q1 to Q10 are General Aptitude, Q1 to Q5 at one mark and Q6 to Q10 at two; the technical questions run from Q11 to Q65, Q11 to Q35 at one mark and Q36 to Q65 at two. A citation such as "2025 (Q35)" in this book is the paper's own printed number.

Where a question could honestly sit in two sections, both are recorded. The question is placed by the boundary rules stated in Section 2.4, the other candidate section is recorded beside it, and where no rule settles the matter the question is marked as counted at its lean, with the reason written out. **23 of the 165 technical questions** carry an alternative, and six of those are counted at a lean. Section 4.7 prints the six and shows what moving them would do.

Why not use a published weightage table. Because a table that does not print its boundary decisions cannot be checked, and in DA the boundaries decide real marks: 34 of the 255 technical marks set so far sit on questions that could have been counted in a second section. A count that names every question it placed and where else it could have gone is a different kind of object from a table of totals. For a paper with three papers in its history, doing the count properly is also not much work, which leaves no reason to rely on anyone else's.

4.2 The papers

FACT.

Year	Organising institute	Session recorded in the key	Question paper	Answer key
2026	IIT Guwahati	Session 8	counted	official key
2025	IIT Roorkee	Session 5	counted	official key
2024	IISc Bengaluru	Session 1	counted	official final key

Three facts about this base that you should have rather than discover:

- **There is nothing before 2024.** DA did not exist as a GATE paper before then. Any "DA previous-year question" dated earlier is a question from some other paper.
- **Each year has one DA paper.** The key records a single session against every DA row, so a year and a question number identify a question completely.
- **One question was awarded to everyone.** The official key marks 2024 (Q59) as MTA, marks to all; the joint density it prints integrates to 4, not 1. It is counted, as the two marks of Probability and Statistics it was set as, because the paper awarded them.

4.3 Marks by section, every paper

FACT. Technical marks by section. General Aptitude is 15 in every paper and is listed last.

Section	2024	2025	2026	Mean	Range	Share of 255 technical marks
1. Probability and Statistics	15	19	21	18.3	15-21	21.6%
4. Programming, Data Structures and Algorithms	20	14	14	16.0	14-20	18.8%
6. Machine Learning	15	12	14	13.7	12-15	16.1%
5. Database Management and Warehousing	7	11	18	12.0	7-18	14.1%
2. Linear Algebra	10	12	8	10.0	8-12	11.8%
7. AI	10	6	8	8.0	6-10	9.4%
3. Calculus and Optimization	8	11	2	7.0	2-11	8.2%
<i>Outside the 2027 syllabus</i>	0	0	0	<i>0.0</i>	0-0	0%
General Aptitude	15	15	15	15.0	15-15	—

FACT. Questions and question types by section, all three papers together:

Section	Questions	One-mark	Two-mark	MCQ	MSQ	NAT
1. Probability and Statistics	34	13	21	16	4	14
2. Linear Algebra	19	8	11	8	8	3

Concept Tested: CUBE as the union of GROUP BY over every subset of its columns, counting only groups that exist.

Why This Question Is High Probability: 2025 (Q46) asked the result size of GROUP BY CUBE with every combination present; this version has missing combinations and duplicate rows.

SOLUTION

CUBE(region, product) produces four groupings, and each contributes one row per distinct group present in the data.

Grouping	Groups	Rows
(region, product)	North-TV, North-Phone, South-TV, East-Laptop, East-Phone, West-TV	6
(region)	North, South, East, West	4
(product)	TV, Phone, Laptop	3
()	grand total	1

Total: $6 + 4 + 3 + 1 = 14$.

Three slips give other numbers. Counting the eight table rows instead of the six distinct pairs gives $8 + 4 + 3 + 1 = 16$. Assuming every region-product pair is present gives $(4 + 1)(3 + 1) = 20$, the 2025 case.

Using ROLLUP (region, product), which omits the product-only grouping, gives $6 + 4 + 1 = 11$.

5.6 Machine Learning

Machine Learning drew **15, 12 and 14 marks** in the three counted DA papers — 2024, 2025 and 2026 — for **41 marks from 27 questions**, a mean of 13.7 marks a paper out of 100. It is one of the heaviest sections in the paper, and it is the one that most resembles the subject the paper is named after.

It is also the section in which the other six meet. A naive Bayes posterior is Bayes' theorem, a principal component is an eigenvector, a gradient step is a derivative and a k-nearest-neighbour vote is a sort.

The counting in this book puts a question here when it asks about the **method** — what the classifier predicts, what the clustering merges, what the network outputs — and elsewhere when the method only supplies a name for a fact that belongs to another section. Every citation below names the year and the question number as printed on the official paper.

The official 2027 syllabus for this section

FACT — GATE 2027 DA, Section 6: Machine Learning (official syllabus PDF, IIT Madras)

Supervised Learning: regression and classification problems, simple linear regression, multiple linear regression, ridge regression, logistic regression, k-nearest neighbour, naive Bayes classifier, linear discriminant analysis, support vector machine, decision trees, bias-variance trade-off, cross-validation methods such as leave-one-out (LOO) cross-validation, k-folds cross-validation, multi-layer perceptron, feed-forward neural network;

Unsupervised Learning: clustering algorithms, k-means/k-medoid, hierarchical clustering, top-down, bottom-up: single-linkage, multiple-linkage, dimensionality reduction, principal component analysis.

A note on that text. "multiple-linkage" is the syllabus's own word and is reproduced as printed; the linkage the papers have actually examined beside single linkage is **complete** linkage (2025 (Q30)). The list names methods but not evaluation metrics, the perceptron rule or gradient descent, all of which the papers have nevertheless asked under "regression and classification problems".

5.6.1 What the 2027 syllabus changed here

FACT. Nothing examinable. The whole DA syllabus has **0 phrases added, 0 removed and 0 reworded** against 2026; in this section the only changes are labels — "(i) Supervised Learning" became "Supervised Learning", "(ii) Unsupervised Learning" became "Unsupervised Learning", and the two are now set as separate paragraphs.

ANALYSIS. So the three counted papers were all set on the words you will be examined on: the 2024 and 2025 syllabi print this section exactly as 2026's does, "(i)" and "(ii)" included. There is no newly named topic to weight up and no dropped one to discount: the evidence below applies to 2027 without adjustment.

5.6.2 What the three papers actually asked

FACT. Every Section 6 row of the three count files.

Year	Questions	Marks	1-mark	2-mark	MCQ	MSQ	NAT
2024	10	15	5	5	7	1	2
2025	8	12	4	4	2	3	3
2026	9	14	4	5	4	2	3
Total	27	41	13	14	13	6	8

FACT — by topic group. The groups are this chapter's, built from the syllabus list; each question is counted once, under the method it asks about.

Topic group	Marks	Questions	2024	2025	2026
Regression and classification problems (framing, separability, training rules, metrics)	9	6	2	3	4
Multi-layer perceptron, feed-forward neural network	6	3	2	2	2
Hierarchical clustering	5	3	2	1	2
Simple, multiple and ridge regression; bias-variance	5	3	0	1	4
Naive Bayes, logistic regression, LDA	4	4	3	1	0
Support vector machine	3	2	1	2	0
Principal component analysis	3	2	0	2	1
k-nearest neighbour	2	1	2	0	0
Decision trees	2	1	2	0	0
Cross-validation	1	1	0	0	1
k-means / k-medoid	1	1	1	0	0

Two 2025 rows were counted at a lean rather than with confidence — Q35 (a naive Bayes misclassification probability that is also plain Bayes arithmetic) and Q60 (PCA variance that is also the largest eigenvalue).

Moving both out would take 2025 to 9 marks.

ANALYSIS — patterns, not trends. Three papers are too few to call anything a trend, and the per-year columns swing hard: regression was 0 marks in 2024 and 4 in 2026. What three papers can show is

recurrence. Three groups appeared **in all three papers** — classification problems, neural networks and hierarchical clustering — and they carry 20 of the 41 marks. Four appeared in **two**: regression, the probabilistic classifiers, SVM and PCA. Four appeared in **one**: k-NN, decision trees, cross-validation and k-means, each in a single question. **None** of the counted papers has examined multiple linear regression, k-fold cross-validation, top-down (divisive) clustering or a logistic-regression calculation, though all four are named. The type mix is the paper's own: MCQ, MSQ and NAT all appear; and two-mark questions are slightly more than half.

3. **Work every calculation by hand on a 4-to-6-point dataset.** The papers set small data precisely so it can be done without a computer; practise that, including stating the tie rule to yourself.
4. **Judgement: cover the named-but-unexamined topics at the level of one calculation each** — multiple linear regression, k-fold counts, logistic $\sigma(w^T x + b)$, divisive clustering. Three papers are too few to treat a zero as a signal.
5. **Keep the concepts that turn up as MSQ statements straight:** generative against discriminative, ridge against lasso, single against complete linkage, what PCA does and does not use.

Question Bank — Machine Learning

Forty questions, written for this book, in rough proportion to the counted evidence — six on regression and classification problems, five on regression, five on the probabilistic classifiers, three each on k-NN, SVM, decision trees, k-means/k-medoid, hierarchical clustering and PCA, two on cross-validation and four on neural networks — with the named-but-unexamined topics given one question each. Every numerical answer was computed, and every stem states the metric, tie rule or convention that makes its answer unique.

QML.1

MCQ · 1 mark

Easy

Subject: Machine Learning | Topic: Regression and classification problems | Subtopic: Telling a regression task from a classification task

Prediction Confidence: Topic 90% | Concept 70% | Question-pattern 55% | Exact-question 12%

Based on PYQs: GATE 2026 (Q23) · last tested 2026 · new question on the same concept, not a reprint

QUESTION

Which one of the following is a **regression** problem?

- (A) Deciding whether an incoming email is spam or not spam (B) Predicting tomorrow's maximum temperature in °C at a weather station (C) Recognising which digit, 0 to 9, a handwritten image shows (D) Deciding which of three species an iris flower belongs to

ANSWER (B)

Concept Tested: A regression target is a continuous quantity; a classification target is one of a finite set of labels, even when the labels are written as numbers.

Why This Question Is High Probability: "regression and classification problems" opens the syllabus list, and 2026 (Q23) asked candidates to match tasks to the algorithms that suit them.

SOLUTION

The test is the **type of the output**, not the type of the input.

(B) TRUE. Temperature is a real number; a prediction of 31.4 °C against a true 31.9 °C is *close*, and the loss (squared or absolute error) measures how close. That is regression.

(A) is binary classification: two labels, spam and not spam.

(C) is the intended trap. The labels are the numerals 0 to 9, but they are **categories**: predicting 7 for a true 8 is no "nearer" than predicting 1. The error made is treating numeric-looking labels as a continuous target.

(D) is three-class classification: the output is a category name.

A useful check: if averaging two outputs gives a meaningful output (the mean of two temperatures is a temperature), it is regression; if not (the mean of "spam" and "not spam" is nothing), it is classification.

QML.2

MSQ · 2 marks

Moderate

Subject: Machine Learning | Topic: Regression and classification problems | Subtopic: Linear separability of two-class datasets

Prediction Confidence: Topic 90% | Concept 72% | Question-pattern 60% | Exact-question 14%

Based on PYQs: GATE 2024 (Q53) · last tested 2024 · new question on the same concept, not a reprint

QUESTION

Each option lists the points of a two-class dataset in $\mathbb{R}^2$. A dataset is **linearly separable** if some line $w_1x_1 + w_2x_2 + b = 0$ has every "+" point strictly on one side and every "-" point strictly on the other. Which of the datasets is/are linearly separable?

- (A) "+": (1, 1); "-": (0, 0), (0, 1), (1, 0)
 (B) "+": (0, 1), (1, 0); "-": (0, 0), (1, 1)
 (C) "+": (0, 0); "-": (2, 0), (-2, 0), (0, 2), (0, -2)
 (D) "+": (1, 1), (2, 3); "-": (3, 1), (4, 2)

ANSWER (A), (D)

Concept Tested: Two finite sets are linearly separable exactly when their convex hulls do not meet.

Why This Question Is High Probability: 2024 (Q53) set this very judgement as a two-mark MSQ on four figured datasets; it is the idea on which the perceptron, logistic regression and SVM all depend.

SOLUTION

Test each set by the convex-hull criterion, and confirm a separating line where one exists.

(A) TRUE — this is AND. The line $x_1 + x_2 = 1.5$ puts (1, 1) at $2 > 1.5$ and the other three at 0 or $1 < 1.5$.

(B) FALSE — this is XOR. The "+" hull is the segment from (0, 1) to (1, 0); the "-" hull is the segment from (0, 0) to (1, 1). The two diagonals cross at (0.5, 0.5), so no line separates them. This is the dataset that a single perceptron cannot learn.

(C) FALSE. The "-" points are the corners of a square (a diamond) around the origin, and (0, 0) lies **inside** their convex hull. Any line with (0, 0) strictly on one side has at least one of the four corners on the same side. The data are separable after a feature map (for example $x_1^2 + x_2^2 < 1$), but not by a line in the original space.

(D) TRUE. The hulls are the segments (1, 1)–(2, 3) and (3, 1)–(4, 2), which do not meet. The line $x_1 - x_2 = 1$ works: the "+" points give 0 and -1 (both < 1), the "-" points give 2 and 2 (both > 1).

The trap is **(C)**: the classes are geometrically well apart, which is not the same as being linearly separable.

QML.3

MCQ · 1 mark

Easy

Subject: Machine Learning | Topic: Regression and classification problems | Subtopic: Perceptron update on one misclassified sample

Prediction Confidence: Topic 90% | Concept 74% | Question-pattern 62% | Exact-question 15%

Based on PYQs: GATE 2025 (Q22) · last tested 2025 · new question on the same concept, not a reprint

QUESTION

A linear classifier predicts $\hat{y} = +1$ if $w^T x + b > 0$ and $\hat{y} = -1$ otherwise. It is trained by the perceptron rule with learning rate $\eta = 0.5$: on a misclassified sample (x, y) , $w \leftarrow w + \eta y x$ and $b \leftarrow b + \eta y$; on a correctly classified sample nothing changes. The current parameters are $w = (1, -2)$, $b = 0$. The sample $x = (2, 1)$ with label $y = +1$ is presented. After this step,

- (A) $w = (1, -2)$, $b = 0$
 (B) $w = (0, -2.5)$, $b = -0.5$
 (C) $w = (2, -1.5)$, $b = 0.5$
 (D) $w = (3, -1)$, $b = 1$

ANSWER (C)

Concept Tested: The perceptron update, and the convention that a score of exactly zero is a prediction of -1 under a strict " > 0 " rule.

Why This Question Is High Probability: 2025 (Q22) set a perceptron-style update of a linear classifier after a misclassified sample; the rule is short and the tie convention is the natural place for a trap.

SOLUTION

Step 1 — the score. $w^T x + b = (1)(2) + (-2)(1) + 0 = 0$.

Step 2 — the prediction. The rule needs a score > 0 for $+1$, so $\hat{y} = -1$. The true label is $+1$: the sample is misclassified.

Step 3 — the update.

$$w \leftarrow (1, -2) + 0.5 \times (+1) \times (2, 1) = (1 + 1, -2 + 0.5) = (2, -1.5) \quad b \leftarrow 0 + 0.5 \times (+1) = 0.5$$

Answer (C). The new score on the same sample is $4 - 1.5 + 0.5 = 3 > 0$, so it is now classified correctly.

(A) $w = (1, -2)$, $b = 0$ treats a zero score as correct (or as "no decision") and skips the update.

(B) subtracts $y \cdot x$ instead of adding it, pushing the score the wrong way.

(D) ignores the learning rate, adding the full $y \cdot x$.

QML.4

MCQ · 2 marks

Moderate

Subject: Machine Learning | Topic: Regression and classification problems | Subtopic: One pass of the perceptron algorithm

Prediction Confidence: Topic 90% | Concept 72% | Question-pattern 58% | Exact-question 12%

Based on PYQs: GATE 2025 (Q22) · last tested 2025 · new question on the same concept, not a reprint

QUESTION

The perceptron of QML.3 (predict $+1$ iff $w^T x + b > 0$, otherwise -1) is trained with learning rate $\eta = 1$, starting from $w = (0, 0)$, $b = 0$. The four samples are presented once, in this order:

$$x_1 = (2, 1), y_1 = +1; x_2 = (1, 3), y_2 = -1; x_3 = (-1, -1), y_3 = -1; x_4 = (3, -1), y_4 = +1.$$

After this single pass, the parameters are

(A) $w = (0, -2)$, $b = -2$

(B) $w = (2, -1)$, $b = 0$

(C) $w = (2, -1)$, $b = -1$

(D) $w = (1, -2)$, $b = 0$

ANSWER (C)

Concept Tested: Sequential perceptron updates, with the zero-score convention applied at every step and the bias updated alongside the weights.

Why This Question Is High Probability: It extends the one-step update of 2025 (Q22) to a short sequence, which is the natural two-mark version.

SOLUTION

Track (w, b) sample by sample.

Step	Sample	Score $w^T x + b$	$\hat{y}$	Mistake?	New w	New b
1	(2, 1), +1	0	-1	yes	(2, 1)	1
2	(1, 3), -1	$2 + 3 + 1 = 6$	+1	yes	(1, -2)	0
3	(-1, -1), -1	$-1 + 2 + 0 = 1$	+1	yes	(2, -1)	-1
4	(3, -1), +1	$6 + 1 - 1 = 6$	+1	no	(2, -1)	-1

Answer (C). A second pass makes no mistakes (scores 2, -2, -2, 6 against labels +, -, -, +), so the algorithm has converged — as the convergence theorem promises for separable data.

Section 9 — The Hundred Questions to Work First

If you have time for a hundred questions and not for 257, these are the hundred. If you have time for all 257, work these first anyway, because finishing them leaves you in a defensible position if the time runs out.

9.1 How the hundred were chosen

This list is not hand-picked. One rule was applied to all 257 questions in the eight subject chapters and the result printed. The rule is stated here so that you can disagree with it precisely.

How many come from each section. Each section's mean marks per paper across the three counted papers, rounded so that the quotas sum to exactly 100:

Section	Mean marks per paper	Questions in the hundred
Probability and Statistics	18.3	18
Programming, Data Structures and Algorithms	16.0	16
General Aptitude	15.0	15
Machine Learning	13.7	14
Database Management and Warehousing	12.0	12
Linear Algebra	10.0	10
AI	8.0	8
Calculus and Optimization	7.0	7

So the hundred mirror the paper. An hour spent on this list is divided the way the marks have been. The means rest on three papers, and a section's marks moved by several between papers, so read each quota as a reasonable share rather than a forecast.

Which questions within a section. Topic confidence highest first — of the four confidence figures this is the one with the most evidence behind it, and Section 7 explains what it is and is not. Ties break on Question-pattern confidence, because a stable question *shape* is a skill that practice transfers to. Any remaining ties break on question number, so the order does not shuffle between printings.

Nothing else enters. Difficulty does not, because an easy question on a near-certain topic is worth more than a hard one on an unlikely topic, and this list is about return on an hour rather than about challenge.

Why the quota is there. Ranked on confidence alone, with no quota, the same rule would have taken 16 from General Aptitude, 15 from AI, 15 from Database Management and Warehousing, 13 from Machine Learning, 13 from Probability and Statistics, 13 from Programming, Data Structures and Algorithms, 10 from Linear Algebra and 5 from Calculus and Optimization. That would have short-changed Probability and Statistics (13 against a quota of 18) — not because the section matters less, but because fewer of its questions carry the very highest Topic figures. A study list that under-weights a section the paper reliably examines is wrong however cleanly it was derived, so the quota decides how many and the confidence figures decide which.

9.2 The blind spot in the rule, stated before the table

The rule ranks on Topic confidence, and Topic confidence rests mostly on the counted record. A topic the syllabus names that none of the three papers has set — confidence intervals and hypothesis tests, LU decomposition, linked lists — has ****no counted history at all**, and so some of the lowest confidence figures in the book**, and its questions drop to the bottom of this list or off it.

That is backwards as a study instruction. A never-examined topic is exactly the material no amount of past-paper practice will make you meet, so it is the material most likely to be missing from your preparation on the day. Section 9.4 lists those questions separately, together with every other question whose concept the counted record has never examined. **Work the hundred, then work Section 9.4.** Do not let the ranking decide that a named topic is unimportant; it decided that only because three papers have not yet set it.

9.3 The hundred

Mk is the question's marks. **T** and **P** are its Topic and Question-pattern confidence. Work the question from the subject chapter, not from this table — the table is an index, and the chapters hold the solutions, the acceptance bands and the reasoning.

#	Question	Section	Topic — subtopic	Type	Mk	T	P
1	QPS.26	Probability and Statistics	Exponential distribution — Memoryless property	MCQ	1	92%	70%
2	QPS.25	Probability and Statistics	Continuous random variables and probability distribution function — Normalising a density	NAT	1	92%	64%
3	QPS.32	Probability and Statistics	Continuous random variables, uniform — Conditional probability for a uniform	NAT	1	92%	62%
4	QPS.28	Probability and Statistics	Cumulative distribution function — Mixed distributions and strict inequalities	MCQ	2	92%	60%
5	QPS.27	Probability and Statistics	Continuous random variables, uniform — Probability of a region for two independent uniforms	NAT	2	92%	58%
6	QPS.29	Probability and Statistics	Cumulative distribution function — Properties of a valid CDF	MSQ	2	92%	58%
7	QPS.30	Probability and Statistics	Exponential distribution — Minimum of independent exponentials	NAT	2	92%	58%
8	QPS.31	Probability and Statistics	Conditional PDF — Conditional mean on a triangular support	MCQ	2	92%	54%
9	QPS.33	Probability and Statistics	Cumulative distribution function, exponential — Distribution of a function of a random variable	MCQ	1	92%	50%
10	QPS.6	Probability and Statistics	Bayes Theorem — Posterior over three sources	NAT	2	88%	68%
11	QPS.8	Probability and Statistics	Bayes Theorem — Updating on two independent tests	MCQ	2	88%	60%
12	QPS.7	Probability and Statistics	Conditional probability — Conditioning on "at least one"	MCQ	1	88%	56%
13	QPS.9	Probability and Statistics	Marginal, conditional and joint probability — Draws without replacement	MSQ	2	88%	54%
14	QPS.35	Probability and Statistics	Normal, standard normal — Standardising and reading the upper tail	MCQ	1	85%	64%
15	QPS.1	Probability and Statistics	Counting (permutation and combinations) — Selection without replacement	MCQ	1	85%	60%
16	QPS.2	Probability and Statistics	Counting (permutation and combinations) — Integer solutions with bounds	NAT	1	85%	58%
17	QPS.34	Probability and Statistics	Poisson distribution — Rescaling the rate to the interval	MCQ	1	85%	58%
18	QPS.36	Probability and Statistics	t-distribution, standard normal — Shape and moments of the t-distribution	MSQ	1	85%	58%
19	QPD.7	Programming, Data Structures and Algorithms	Programming in Python — Tracing a recursive function	NAT	2	92%	74%
20	QPD.8	Programming, Data Structures and Algorithms	Programming in Python — Counting recursive calls	MCQ	2	92%	70%
21	QPD.1	Programming, Data Structures and Algorithms	Programming in Python — Aliasing: += against + on lists	MCQ	1	92%	68%
22	QPD.2	Programming, Data Structures and Algorithms	Programming in Python — Mutable default argument	MCQ	1	92%	66%
23	QPD.10	Programming, Data Structures and Algorithms	Programming in Python — Recursion over a tree stored in a dictionary	MCQ	2	92%	66%
24	QPD.12	Programming, Data Structures and Algorithms	Programming in Python — Tracing an array loop	NAT	2	92%	66%
25	QPD.6	Programming, Data Structures and Algorithms	Programming in Python — Set operations inside a loop	MCQ	2	92%	64%
26	QPD.11	Programming, Data Structures and Algorithms	Programming in Python — What a recursive function computes	MCQ	2	92%	64%
27	QPD.9	Programming, Data Structures and Algorithms	Programming in Python — Closures and late binding	MSQ	2	92%	62%
28	QPD.3	Programming, Data Structures and Algorithms	Programming in Python — List replication and shared rows	MCQ	1	92%	60%
29	QPD.4	Programming, Data Structures and Algorithms	Programming in Python — Floor division, modulo and rounding	MSQ	1	92%	58%
30	QPD.5	Programming, Data Structures and Algorithms	Programming in Python — Local and global scope	MCQ	1	92%	54%
31	QPD.13	Programming, Data Structures and Algorithms	Programming in Python — Nested loops with break and for ... else	NAT	2	92%	54%
32	QPD.27	Programming, Data Structures and Algorithms	Basic sorting algorithms — Selection sort, state after k passes	MCQ	1	86%	70%
33	QPD.28	Programming, Data Structures and Algorithms	Basic sorting algorithms — Insertion sort swaps and inversions	MSQ	1	86%	70%
34	QPD.29	Programming, Data Structures and Algorithms	Basic sorting algorithms — Bubble sort with early exit — comparisons	NAT	2	86%	70%
35	QGA.8	General Aptitude	Quantitative Aptitude — Percentage change in mensuration	MCQ	1	92%	70%
36	QGA.5	General Aptitude	Quantitative Aptitude — Counting with a divisibility rule	MCQ	1	92%	68%
37	QGA.7	General Aptitude	Quantitative Aptitude — Conditional probability	MCQ	2	92%	66%

Section 10 — What Not to Skip, and What to Skip

There are two ways to be wrong about a syllabus. You can study something it does not contain, which costs time. You can skip something it does contain, which costs marks. **These are not symmetric**, and treating them as though they were is the mistake behind most reading lists.

The asymmetry has a practical consequence that runs through this whole chapter: where a topic's status is genuinely unclear, read it briefly rather than skip it. An hour lost is recoverable. A question you cannot attempt is not.

This chapter gives two lists of what not to skip, built from opposite kinds of evidence, and then says what can be skipped. The last answer is shorter than in most books of this kind, for a reason Section 10.3 gives.

10.1 Do not skip, group one: the topics examined in all three counted papers

The strongest signal three papers can give is a topic that drew marks in **every one of them** — 2024, 2025 and 2026, three different setting institutes, no gap. Three papers show patterns, not trends, and this is the clearest pattern they show: a topic the examiners have not once left out.

25 of the 57 topic groups in the counted record did so. They carry 195 of the 300 marks counted across the three papers. If your preparation is incomplete, it should be incomplete somewhere other than here.

Topic	Section	Counted marks	Papers
Programming in Python (code and pseudocode traces)	Programming, Data Structures and Algorithms	18	3 of 3
Quantitative aptitude: computation, geometry, statistics, probability	General Aptitude	15	3 of 3
Continuous random variables: uniform, exponential, CDF	Probability and Statistics	13	3 of 3
Logic: propositional and predicate	AI	10	3 of 3
Search: uninformed, informed, adversarial	AI	10	3 of 3
Spatial aptitude	General Aptitude	9	3 of 3
Graph theory and graph algorithms: traversals, shortest path	Programming, Data Structures and Algorithms	9	3 of 3
Maxima and minima, optimisation of a single variable	Calculus and Optimization	8	3 of 3
Relational algebra and tuple calculus	Database Management and Warehousing	8	3 of 3
Analytical aptitude	General Aptitude	8	3 of 3
Verbal aptitude	General Aptitude	8	3 of 3
Classification problems and linear discriminants	Machine Learning	8	3 of 3
Projection, orthogonal, idempotent and partition matrices	Linear Algebra	7	3 of 3
Poisson, normal, standard normal, t and chi-squared distributions	Probability and Statistics	7	3 of 3
Normal form and functional dependencies	Database Management and Warehousing	6	3 of 3
SQL	Database Management and Warehousing	6	3 of 3
Clustering: k-means, k-medoid, hierarchical	Machine Learning	6	3 of 3
Multi-layer perceptron and feed-forward neural networks	Machine Learning	6	3 of 3
Counting, probability axioms, sample space and events	Probability and Statistics	6	3 of 3

Topic	Section	Counted marks	Papers
Eigenvalues, eigenvectors and determinant	Linear Algebra	5	3 of 3
Conditional and joint probability, Bayes' theorem	Probability and Statistics	5	3 of 3
Basic sorting algorithms: selection, bubble, insertion	Programming, Data Structures and Algorithms	5	3 of 3
Limit	Calculus and Optimization	4	3 of 3
Vector space, subspaces, linear dependence and independence	Linear Algebra	4	3 of 3
Search algorithms: linear and binary search	Programming, Data Structures and Algorithms	4	3 of 3

A caution about the topics *not* in that table

****A low paper-count is much weaker evidence than a high one, and this book draws no skip conclusion from it.****
There are two reasons.

The first is the size of the record. With three papers, a topic set once is as consistent with *rare* as with *not drawn yet*, and nothing in the record can tell the two apart.

The second is mechanical. Every counted question is placed in exactly one section and one topic, and DA's sections overlap by design, so a skill used inside questions counted elsewhere shows a low count of its own without being rare in the examination. The sharpest case in the record is Calculus and Optimization in 2026: the section was counted at **2 marks**, while 5 more marks in the same paper — 2026 (Q29), 2026 (Q45) and 2026 (Q65) — used it as the working tool and were counted where their method sits. A reader of the section total alone would conclude that the paper hardly asked for it. It asked for it more than the total shows.

Read the table above as *these are certain*. Do not read its absences as *these are safe*.

10.2 Do not skip, group two: what the syllabus names and no paper has set

These have no past-paper history at all, so ****no amount of previous-year practice will make you meet them****. That is exactly why they belong on a do-not-skip list: they are the material most likely to be missing from a preparation built on past papers.

First, the whole topic groups that have drawn no marks in the three papers:

Section	Topic named in the syllabus, with no counted marks
Probability and Statistics	Confidence interval, z-test, t-test, chi-squared test
Linear Algebra	LU decomposition
Programming, Data Structures and Algorithms	Linked lists

Second, the named items inside topic groups that have drawn marks, where no counted question's recorded topic names the item itself. These are finer-grained, and a question on a neighbouring item may have touched one in passing, so read this list as *not yet set as a question in its own right*:

Section	Named in the syllabus, not yet set on its own
Linear Algebra	partition matrix
Programming, Data Structures and Algorithms	linear search
Programming, Data Structures and Algorithms	mergesort
Database Management and Warehousing	file organization, as a topic of its own

Section 11 — The Ten Mock Tests

Ten full papers follow. Each is 65 questions, 100 marks and three hours, built to the GATE Data Science and Artificial Intelligence pattern and to the section weights of the three counted papers. Every question in them was written for this book; none reproduces a past paper and none repeats a question from the subject chapters, which is why no mock question carries a year as a source.

A mock test is not a question bank with a clock attached. It measures three things the bank cannot: whether you recognise a topic when nothing announces it, whether you can hold accuracy for three hours, and where your time actually goes as opposed to where you think it goes. Use them for that.

The common blueprint

All ten papers are laid out identically, because the real paper is: the 2024, 2025 and 2026 papers share this layout exactly.

Questions	Block	Marks each
Q1 – Q5	General Aptitude	1
Q6 – Q10	General Aptitude	2
Q11 – Q35	Technical	1
Q36 – Q65	Technical	2

That gives 30 one-mark and 35 two-mark questions, General Aptitude at exactly 15 marks and the technical sections at 85 — the shape all three counted papers close on. In all three, every General Aptitude question was a multiple-choice question, and the mocks keep that too.

The 85 technical marks are distributed as the three counted papers distribute them. The target for each section is its three-paper mean rounded to whole marks:

Section	2024	2025	2026	Mean	Target in each mock
Probability and Statistics	15	19	21	18.3	18
Programming, Data Structures and Algorithms	20	14	14	16.0	16
Machine Learning	15	12	14	13.7	14
Database Management and Warehousing	7	11	18	12.0	12
Linear Algebra	10	12	8	10.0	10
AI	10	6	8	8.0	8
Calculus and Optimization	8	11	2	7.0	7

Two consequences are worth saying out loud. **The mocks vary the weights a little, but far less than the real paper has.** A mock may move a section by a mark or two from its target, as a real paper does, and never by more than three. The three real papers moved much further: Database Management and Warehousing ran from 7 marks to 18, and Calculus and Optimization from 11 marks to 2. A practice paper that swung that far would mix two measurements at once — your preparation and the paper's shape — so the mocks do not, but it means no mock can show you how a two-mark Calculus year feels. **And nothing in the mocks has to be set aside for syllabus changes**, because there were none: the 2027 syllabus text is word for word the text all three counted papers were set on.

Marking, exactly as the official keys apply it.

Type	Correct	Wrong	Partly correct
MCQ	full marks	minus one-third of the question's marks	not possible
MSQ	full marks	no penalty	no credit
NAT	full marks for any value in the band	no penalty	not applicable

So a wrong one-mark MCQ costs $1/3$ and a wrong two-mark MCQ costs $2/3$. MSQ scores only when every correct option is selected and no incorrect one is — no partial credit, no negative marking — which makes it the format that punishes partial knowledge hardest. NAT carries no penalty at all, so a NAT left blank is strictly worse than a NAT guessed.

Each paper states its own conventions — that code is Python 3, how lists and arrays are indexed, and the base of any logarithm — and every numerical-answer question prints its acceptance band. Where a question depends on a tie-breaking rule, an order of expansion or a split convention, the stem states it, because without it the answer is not unique.

Why the MSQ and NAT counts vary between papers

The number of multiple-select and numerical-answer questions is different in each of the ten papers, and that is deliberate, because it is different in each real paper. Counted from the official answer keys:

Paper	MCQ	MSQ	NAT
2024	37	12	16
2025	35	18	12
2026	33	14	18

Because General Aptitude has been all MCQ, every MSQ and every NAT in those papers sat in the technical 85 marks: between 12 and 18 of each in a paper. The mocks stay inside that range and move within it from paper to paper. A set of mocks with a constant MSQ count would train you for a paper that does not exist.

Varying it means some of these ten papers will feel harsher than others for reasons that have nothing to do with your preparation — which is exactly what the record supports. A paper heavy in MSQ is the harsher one: eighteen questions on which a single wrong tick scores nothing.

Scoring a paper, and what to do with the score

Sit the paper in one three-hour block. Mark it from Section 12, which gives the key, the type and the section for every question, and read the full working in Section 13.

Then compare section by section, not total by total. Each mock question is tagged with its section, so a paper resolves into eight scores as easily as one. A total moves for too many reasons at once — a hard Machine Learning block, a good day on Aptitude, a lucky MSQ — and comparing totals across papers tells you almost nothing you can act on. Comparing your Database score across four papers tells you whether Database is improving. Keep one table, eight rows, one column per paper; and the diagnosis reads itself.

And classify every lost mark by cause, because the four causes need four different remedies.

Cause	What it looks like
Misread	You solved a different question — missed "not", "with replacement", "at most", "sample" for population, or "0-indexed"
Wrong formula	Right topic, right method, wrong expression — $\text{Var}(aX)$ as $a\text{Var}(X)$, n where $n - 1$ belongs, entropy in natural logs where bits were asked

Mock Test 01

Duration: 180 minutes | Maximum marks: 100 | 65 questions

Q1–Q5 General Aptitude at 1 mark, Q6–Q10 General Aptitude at 2 marks, Q11–Q35 technical at 1 mark, Q36–Q65 technical at 2 marks. Inside each technical block the questions run MCQ first, then MSQ, then NAT, which is the order the DA papers use. The technical marks sit exactly on the section targets: Probability and Statistics 18, Programming, Data Structures and Algorithms 16, Machine Learning 14, Database Management and Warehousing 12, Linear Algebra 10, AI 8, Calculus and Optimization 7.

This paper carries fourteen multiple-select and fifteen numerical-answer questions, with thirty-six MCQs. The three counted DA papers ran from 12 to 18 MSQs and from 12 to 18 NATs, so this is a middle-of-the-range paper. All ten General Aptitude questions are MCQ, as they have been in every DA paper so far.

Marking. MCQ: correct +marks, wrong $-1/3$ of marks. MSQ: all correct options and no incorrect one, or nothing; no negative marking. NAT: no negative marking; any value in the printed band scores.

Conventions. "log" is base e unless a base is written. Code is Python 3. Lists and arrays are 0-indexed. SQL follows standard semantics: a comparison with NULL is unknown, and aggregates other than COUNT(*) skip NULLs. The height of a tree counts edges, so a single node has height 0. Adjacency lists are processed in the order written.

Q1 – Q5 carry ONE mark each.

Q1

MCQ · 1 mark

Select the word that completes the analogy. Frugal : Spendthrift :: Candid : _____

- | | |
|----------------------------------|--------------------------------|
| ☐ A Evasive | ☐ B Honest |
| ☐ C Generous | ☐ D Timid |

General Aptitude — verbal analogy

Q2

MCQ · 1 mark

Select the option that fills the blank correctly. "Neither the manager nor the analysts _____ aware of the change in the reporting format."

- | | |
|-----------------------------|----------------------------------|
| ☐ A was | ☐ B were |
| ☐ C is | ☐ D has been |

General Aptitude — subject-verb agreement

Q3

MCQ · 1 mark

The price of an item is raised by 20 per cent and the new price is then reduced by 20 per cent. The net change in the price is

- | | |
|--|--|
| ☐ A a fall of 4 per cent | ☐ B no change |
| ☐ C a rise of 4 per cent | ☐ D a fall of 2 per cent |

General Aptitude — percentages

Q4 MCQ · 1 mark

A wooden cube is painted on all six faces and then cut into 125 identical smaller cubes. The number of smaller cubes with paint on exactly two faces is

- (A) 8 (B) 27
(C) 36 (D) 54

General Aptitude — spatial reasoning

Q5 MCQ · 1 mark

Statements: All data scientists are programmers. Some programmers are musicians. Which one of the following is necessarily true?

- (A) Some data scientists are musicians. (B) All programmers are data scientists.
(C) No data scientist is a musician. (D) Some musicians are programmers.

General Aptitude — syllogism

Q6 – Q10 carry TWO marks each.

Q6 MCQ · 2 marks

P alone can finish a job in 12 days and Q alone in 18 days. They work together for 4 days, after which P leaves. The number of further days Q needs to finish the job is

- (A) 3.2 (B) 8
(C) 12 (D) 14

General Aptitude — time and work

Q7 MCQ · 2 marks

The table gives a company's sales (in units) of two products over four quarters.

Product	Quarter 1	Quarter 2	Quarter 3	Quarter 4
X	100	120	140	170
Y	60	80	100	120

The quarter in which the combined sales of X and Y grew by the largest percentage over the previous quarter is

- (A) Quarter 2
(B) Quarter 3
(C) Quarter 4
(D) the growth was the same in Quarters 2, 3 and 4

General Aptitude — data interpretation

Q8 MCQ · 2 marks

Five people P, Q, R, S and T sit in a row, all facing north. R sits in the middle seat. P sits at the extreme left. T sits immediately to the right of R. Q does not sit next to P. The person at the extreme right is

- (A) S
(B) Q

Q65

NAT · 2 marks

The maximum value of $f(x) = x \cdot e^{(-x/2)}$ over $x \geq 0$ is ____ (round off to three decimal places).

Acceptance band: 0.73 to 0.74.

Calculus and Optimization — optimization in one variable

Answer key — Mock Test 01

Attempt the whole paper before reading this page. Score with the official scheme: a correct 1-mark question gives +1 and a correct 2-mark question +2; a wrong MCQ costs 1/3 of a mark (1-mark) or 2/3 of a mark (2-mark); MSQ and NAT carry no negative marking, and an MSQ scores only if every correct option is selected and no incorrect one is. NAT answers are given to the precision stated in each question.

This paper contains 36 MCQ, 14 MSQ and 15 NAT questions (30 one-mark and 35 two-mark, 100 marks).

Q	Answer	Type	Q	Answer	Type	Q	Answer	Type
1	A	MCQ	23	D	MCQ	45	A	MCQ
2	B	MCQ	24	B;C;D	MSQ	46	D	MCQ
3	A	MCQ	25	A;B;D	MSQ	47	C	MCQ
4	C	MCQ	26	B;D	MSQ	48	D	MCQ
5	D	MCQ	27	A;C;D	MSQ	49	A;C;D	MSQ
6	B	MCQ	28	A;C	MSQ	50	B;C;D	MSQ
7	A	MCQ	29	B;D	MSQ	51	A;B;D	MSQ
8	B	MCQ	30	0.731	NAT	52	A;B	MSQ
9	C	MCQ	31	5	NAT	53	A;B;C	MSQ
10	C	MCQ	32	-0.24	NAT	54	C;D	MSQ
11	C	MCQ	33	1231	NAT	55	B	MSQ
12	A	MCQ	34	30	NAT	56	A;B;C;D	MSQ
13	A	MCQ	35	0.333	NAT	57	1.80	NAT
14	A	MCQ	36	A	MCQ	58	0.213	NAT
15	C	MCQ	37	D	MCQ	59	0.391	NAT
16	C	MCQ	38	A	MCQ	60	23	NAT
17	B	MCQ	39	D	MCQ	61	173	NAT
18	D	MCQ	40	B	MCQ	62	1.604	NAT
19	B	MCQ	41	C	MCQ	63	0.345	NAT
20	B	MCQ	42	B	MCQ	64	9	NAT
21	D	MCQ	43	B	MCQ	65	0.736	NAT
22	C	MCQ	44	C	MCQ			

Detailed solutions for this test: Section 13, under the heading "Mock Test 01".

Each solution there shows the full working and ends with the four confidence scores (Topic, Concept, Question-pattern, Exact-question) for the concept it tests.

Consolidated keys for all 10 tests: Section 12.

Section 13 — Detailed Mock Test Solutions

Each test's solutions follow, in question order. Attempt and score the paper (Section 12) before reading these.

Mock Test 01

Q1

ANSWER A

Confidence: Pattern 70% | Exact-question 12%

A frugal person is careful with money; a spendthrift wastes it. The pair are **opposites**. A candid person speaks frankly, and the opposite of frank is **evasive**.

B. Honest is close to candid in meaning, so it turns the opposite relation into a synonym relation.

C. Generous carries over the money theme of the first pair instead of its relation. **D. Timid** is a contrasting personal trait, but the opposite of timid is bold, not candid.

Q2

ANSWER B

Confidence: Pattern 66% | Exact-question 11%

With *neither ... nor* (and *either ... or*), the verb agrees with the **nearer** subject. The nearer subject is *the analysts*, which is plural, so the verb is **were**.

A. was agrees with the farther subject, *the manager*. **C. is** is singular and also in the wrong tense for a completed change. **D. has been** is singular as well.

Q3

ANSWER A

Confidence: Pattern 70% | Exact-question 12%

Successive changes multiply: $1.20 \times 0.80 = 0.96$. The price ends at 96 per cent of the original, a ****fall of 4 per cent****.

B. assumes a rise and a fall of the same percentage cancel, but the fall is taken on the larger, raised price. **C.** gets the size right and the direction wrong. **D.** halves the true change.

Q4

ANSWER C

Confidence: Pattern 64% | Exact-question 11%

The cube is 5 small cubes along each edge. A small cube has exactly two painted faces when it lies on an edge of the big cube but is not a corner. Each edge has $5 - 2 = 3$ such cubes, and a cube has 12 edges:

$12 \times 3 = 36$.

A. 8 is the number of corner cubes, which have three painted faces. **B. $27 = 3^3$** is the number of interior cubes with no paint. **D. $54 = 6 \times 3^2$** is the number of cubes with exactly one painted face.

Q5

ANSWER D

Confidence: Pattern 66% | Exact-question 11%

"Some programmers are musicians" says at least one person is both a programmer and a musician. That same person shows that **some musicians are programmers**; a particular affirmative statement converts.

A. does not follow: the musicians among the programmers may all be outside the data scientists (the middle term *programmers* is never distributed). **B.** reverses "all data scientists are programmers", which does not convert. **C.** is not supported either; nothing rules out a data scientist who is a musician.

Q6 **ANSWER B**

Confidence: Pattern 70% | Exact-question 12%

Working together, P and Q do $1/12 + 1/18 = 5/36$ of the job per day. In 4 days they do $20/36 = 5/9$, leaving $4/9$. Q alone takes $(4/9) \times 18 = 8$ days.

A. 3.2 finishes the remaining $4/9$ at the combined rate, $(4/9) \div (5/36) = 3.2$, forgetting that P has left.

C. 12 forgets that Q worked during the first 4 days: P alone would do $4/12 = 1/3$, leaving $2/3$ for Q, and $(2/3) \times 18 = 12$. **D. 14** subtracts days, $18 - 4$, as though days of work were interchangeable between workers.

Q7 **ANSWER A**

Confidence: Pattern 68% | Exact-question 11%

Combined sales are 160, 200, 240 and 290. The growth over the previous quarter is

- Quarter 2: $40/160 = 25$ per cent
- Quarter 3: $40/200 = 20$ per cent
- Quarter 4: $50/240 \approx 20.8$ per cent

C. Quarter 4 has the largest **absolute** rise (50), which is not what is asked. **B. Quarter 3** has the same absolute rise as Quarter 2 (40), but on a larger base of 200, so its percentage growth is smaller. **D.** would need the three growth rates to be equal; they are not.

Q8 **ANSWER B**

Confidence: Pattern 68% | Exact-question 11%

Number the seats 1 to 5 from the left. P is in seat 1, R in seat 3 and T in seat 4. Seats 2 and 5 remain for Q and S. Q may not sit next to P, so Q cannot take seat 2; Q takes seat 5 and S seat 2. The person at the extreme right is **Q**.

A. S puts Q in seat 2 by ignoring the last condition. **C. T** reads "immediately to the right of R" as "at the right end". **D.** overlooks that the last condition settles the only remaining choice.

Q9 **ANSWER C**

Confidence: Pattern 66% | Exact-question 11%

There are 36 equally likely ordered outcomes. The prime sums and their counts are 2 (1 way), 3 (2), 5 (4), 7 (6) and 11 (2), a total of 15. The probability is $15/36 = 5/12$.

A. 5/11 treats the eleven possible sums 2 to 12 as equally likely. **B. 7/18** = $14/36$ forgets that 2 is prime. **D. 8/21** counts unordered pairs (21 of them) as equally likely, but (1, 2) and (2, 1) are two different outcomes.

Q10 **ANSWER C**

Confidence: Pattern 64% | Exact-question 11%

The passage reports that two things happened after the charge was introduced; it never says why. A faster bus journey **fits** lighter traffic, but other causes are not ruled out. Only **C** claims no more than the passage supports.

A. turns "after" into "because of", which is a causal claim the passage does not make. **B.** is an extreme claim with no support; a 15 per cent faster bus journey does not require zero cars. **D.** brings in a fact the passage never mentions.

Q11 **ANSWER C**

Confidence: Pattern 72% | Exact-question 13%

With $k = 1$ the prediction follows every training point, noise included: low bias, high variance. As k grows, each prediction averages over more neighbours, so it becomes smoother and more stable (**variance falls**) but less able to follow the true boundary (**bias rises**). At $k = n$ every query gets the majority class.

Section 15 — The Last-Minute Sheet

What follows is the set of formulae, complexities and facts worth carrying into the last day, grouped by the seven technical sections of the 2027 syllabus. Each group opens with the syllabus's own text for that section, quoted as printed, so that the sheet can be checked against what the paper may ask: every topic the syllabus names has an entry here, including the ones no counted paper has yet examined. Every numerical example on the sheet was recomputed before printing.

Two things this sheet deliberately is not. **It is not a syllabus course.** A formula list cannot tell you what a formula is for, and a candidate meeting one of these for the first time on this sheet is meeting it too late. **And it does not replace the chapters.** Each entry is the short form of something the subject chapter works through in full; where a result has a condition — independence, a finite variance, a non-negative edge weight, an admissible heuristic — the condition is printed beside it, because a result used without its condition is the commonest wrong answer on the paper.

Notation: "log" with no base is the natural logarithm and is written $\ln$; $\log_2$ is written out wherever bits are meant. $P(A | B)$ is the probability of A given B. Vectors are columns; x^T is the transpose. n is the number of observations or elements unless stated.

1 Probability and Statistics

Syllabus. Counting (permutation and combinations), probability axioms, Sample space, events, independent events, mutually exclusive events, marginal, conditional and joint probability, Bayes Theorem, conditional expectation and variance, mean, median, mode and standard deviation, correlation, and covariance, random variables, discrete random variables and probability mass functions, uniform, Bernoulli, binomial distribution, Continuous random variables and probability distribution function, uniform, exponential, Poisson, normal, standard normal, t-distribution, chi-squared distributions, cumulative distribution function, Conditional PDF, Central limit theorem, confidence interval, z-test, t-test, chi-squared test.

Counting and probability rules

Item	Result
Permutations, combinations	${}^nP_r = n!/(n-r)!$, ${}^nC_r = n!/(r!(n-r)!)$
Arrangements with repeats	$n!/(n_1! n_2! \dots n_k!)$
Non-negative integer solutions of $x_1 + \dots + x_k = n$	$C(n+k-1, k-1)$; positive solutions $C(n-1, k-1)$. For $x_1 + x_2 + x_3 = 10$: 66 and 36
Addition rule	$P(A \cup B) = P(A) + P(B) - P(A \cap B)$
Complement	$P(\text{at least one}) = 1 - P(\text{none})$
Mutually exclusive	$P(A \cap B) = 0$
Independent	$P(A \cap B) = P(A)P(B)$. Two events with non-zero probabilities cannot be both independent and mutually exclusive
Conditional	$P(A B) = P(A \cap B)/P(B)$, $P(B) > 0$
Total probability	$P(B) = \sum_i P(B A_i)P(A_i)$, the A_i a partition
Bayes' theorem	$P(A_i B) = P(B A_i)P(A_i) / \sum_j P(B A_j)P(A_j)$
Marginal from joint	$P(X = x) = \sum_y P(X = x, Y = y)$; $f_X(x) = \int f(x, y) dy$

Expectation, variance and their relatives

Item	Result
Linearity	$E[aX + bY + c] = aE[X] + bE[Y] + c$ — no independence needed
Variance	$\text{Var } X = E[X^2] - (E[X])^2$; $\text{Var}(aX + b) = a^2 \text{Var } X$
Sum	$\text{Var}(X + Y) = \text{Var } X + \text{Var } Y + 2 \text{Cov}(X, Y)$; the covariance term vanishes only if uncorrelated
Covariance	$\text{Cov}(X, Y) = E[XY] - E[X]E[Y]$; $\text{Cov}(aX, bY) = ab \text{Cov}(X, Y)$
Correlation	$\rho = \text{Cov}(X, Y)/(\sigma_X \sigma_Y)$, $-1 \leq \rho \leq 1$, unchanged by positive rescaling
Independence	independent $\Rightarrow$ uncorrelated; uncorrelated does not imply independent (X uniform on $(-1, 1)$, $Y = X^2$ has $\rho = 0$)
Product, independent	$E[XY] = E[X]E[Y]$
Law of total expectation	$E[X] = E[E[X Y]]$
Law of total variance	$\text{Var } X = E[\text{Var}(X Y)] + \text{Var}(E[X Y])$
Conditional PDF	$f(y x) = f(x, y)/f_X(x)$; $E[Y X = x] = \int y f(y x) dy$
Sample mean, variance	$\bar{x} = \sum x_i/n$; $s^2 = \sum (x_i - \bar{x})^2/(n - 1)$ is the unbiased sample variance; divide by n for the population form
Pairwise-difference identity	$\sum \text{over pairs } i < j \text{ of } (x_i - x_j)^2 = n \sum (x_i - \bar{x})^2$
Adding one point	new mean $= (n \bar{x} + x_{\text{new}})/(n + 1)$
Median	continuous: the m with $F(m) = 0.5$; data: middle value of the sorted list, mean of the two middle values for even n
Mode	most frequent value; for a density, the maximiser of f
Standard deviation	$\sqrt{\text{variance}}$; same units as the data

Distributions

Distribution	PMF / PDF	Mean	Variance
Bernoulli(p)	$P(1) = p, P(0) = 1 - p$	p	$p(1 - p)$
Binomial(n, p)	$C(n, k) p^k (1 - p)^{n-k}$	np	$np(1 - p)$
Discrete uniform on $\{1, \dots, n\}$	$1/n$	$(n + 1)/2$	$(n^2 - 1)/12$
Geometric (trial of first success)	$(1 - p)^{k-1} p, k \geq 1$	$1/p$	$(1 - p)/p^2$
Poisson(λ)	$e^{-\lambda} \lambda^k/k!$	λ	λ
Continuous uniform(a, b)	$1/(b - a)$	$(a + b)/2$	$(b - a)^2/12$
Exponential(λ)	$\lambda e^{-\lambda x}, x \geq 0$	$1/\lambda$	$1/\lambda^2$
Normal(μ, σ^2)	$(1/(\sigma\sqrt{2\pi})) e^{-(x - \mu)^2/(2\sigma^2)}$	μ	σ^2
Chi-squared, k degrees of freedom	sum of k squared independent $N(0, 1)$	k	$2k$
t, v degrees of freedom	symmetric about 0, heavier tails than $N(0, 1)$	$0 (v > 1)$	$v/(v - 2) (v > 2)$

Fact	Result
Exponential tail and CDF	$P(X > x) = e^{-\lambda x}$; $F(x) = 1 - e^{-\lambda x}$; median $\ln 2/\lambda$
Memoryless	$P(X > s + t X > s) = P(X > t)$ — exponential (continuous) and geometric (discrete) only
Floor of an exponential	$[X]$ is geometric on $\{0, 1, 2, \dots\}$: $P([X] = k) = (1 - q)q^k$ with $q = e^{-\lambda}$
Poisson approximation	$\text{Binomial}(n, p) \approx \text{Poisson}(np)$ for large n and small p
Sums	independent Poissons add (rates add); independent normals add (means add, variances add)

Section 16 — Sources, Conventions and Limits

This chapter says what the book was built from, how the counting was done, and what the book does not claim. It is here so that a reader who doubts a figure knows which file to open.

What the book was built from

Two official syllabus documents. The GATE 2026 Data Science and Artificial Intelligence syllabus, published by IIT Guwahati, and the GATE 2027 syllabus, published by IIT Madras, both held as PDFs in `research/official/`. Section 2 reproduces the 2027 document and compares the two word by word; the comparison is produced by a program that reads the two PDFs, not from memory, and it finds no phrase added, removed or reworded — only the numbering of the sections and the labelling of the two Machine Learning paragraphs changed. The 2024 (IISc Bengaluru) and 2025 (IIT Roorkee) DA syllabi, also held in `research/official/`, were compared with 2026's the same way: they name the same phrases word for word, and differ from it by one dash in 2024 and not at all in 2025.

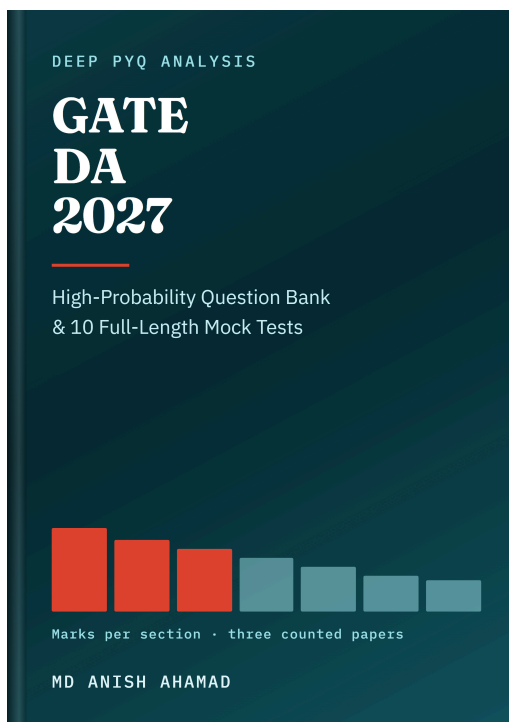
Three official question papers and their three official answer keys — every GATE DA paper there has been. DA was first set in GATE 2024, so the record is the 2024 paper (organised by IISc Bengaluru), the 2025 paper (IIT Roorkee) and the 2026 paper (IIT Guwahati): 195 questions and 300 marks. DA is set as one paper a year, so a citation in this book names a year and the printed question number, as in "2025 (Q35)". Every file was downloaded from a GATE organising-institute website. No coaching site, aggregator, question repository or blog was used for any paper, key, syllabus or figure in this book.

File	Source
2024 question paper	https://gate2026.iitg.ac.in/doc/download/2024/DA24S1.pdf
2024 answer key	https://gate2026.iitg.ac.in/doc/download/2024/DAFinalAnswerKey.pdf
2025 question paper	https://gate2025.iitr.ac.in/doc/2025/2025_QP/DA.pdf
2025 answer key	https://gate2025.iitr.ac.in/doc/2025/2025_Key/DA_Keys.pdf
2026 question paper	https://gate2026.iitg.ac.in/doc/download/2026/QPs/DA.pdf
2026 answer key	https://gate2026.iitg.ac.in/doc/download/2026/Keys/DA_Keys.pdf
2024 syllabus	https://gate2024.iisc.ac.in/wp-content/uploads/2023/08/GATE2024DataScienceAlsyllabus.pdf
2025 syllabus	https://gate2025.iitr.ac.in/doc/2025/GATE%20_DA_2025_Syllabus.pdf
2026 syllabus	https://gate2026.iitg.ac.in/doc/GATE2026_Syllabus/DA_2026_Syllabus.pdf
2027 syllabus	https://gate2027.iitm.ac.in/static/doc/GATE2027_Syllabus/DA_GATE2027_Syllabus.pdf

All ten were downloaded on 30 September 2026. The 2024 paper and key were downloaded from the IIT Guwahati site; the 2024 examination itself was organised by IISc Bengaluru.

****The counting files in `research/`.** ** One JSON file and one readable Markdown file per paper — `count_2024`, `count_2025` and `count_2026` — plus `COUNT_SPEC.md`, the rules every count followed, and the generated syllabus files `syllabus_2027.md` and `syllabus_changes.md`. The syllabus files are generated from the PDFs by script rather than edited by hand, and each paragraph of the reproduced syllabus is checked mechanically against the PDF's own text.

Two things about the source base a reader should have rather than discover. All three papers carry a text layer, and it was the primary source; but a text layer loses figures, tables and some formulas, so the pages that depend on them — game trees, a B+ tree, functional-dependency arrows, concept hierarchies, a pixel grid, network and dataset figures,



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